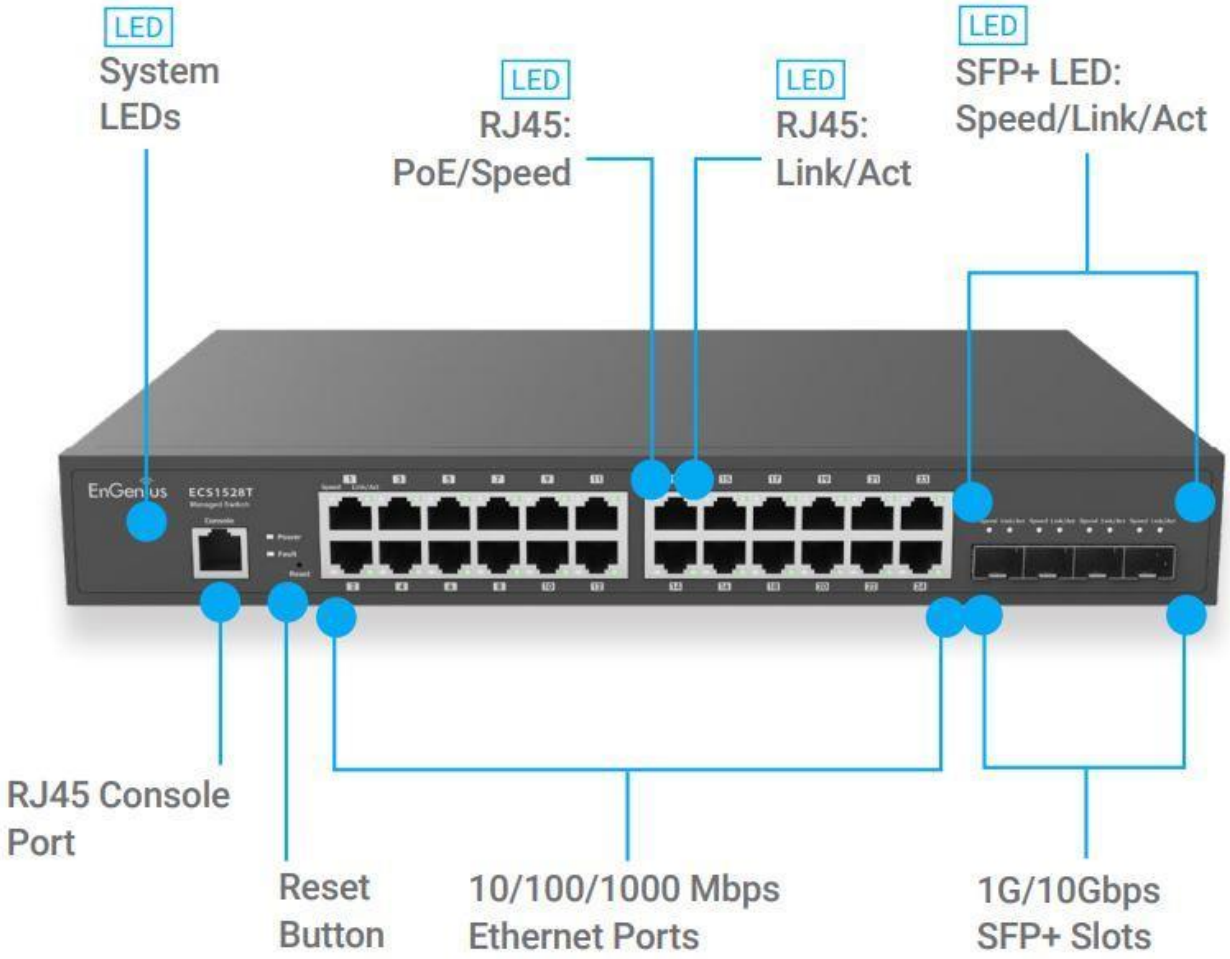
1 Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

1. Network Devices: Functions and Ports

Device	Main Function	Typical Ports
Switch	Connects multiple wired devices in a LAN.	Ethernet ports, Console port, Power port
Router	Connects different networks and provides internet access.	WAN, LAN Ethernet, Power, sometimes USB
Hub	Connects multiple devices and broadcasts data to all ports.	Ethernet ports, Power port
Firewall	Protects the network by filtering incoming and outgoing traffic.	Ethernet/WAN/LAN ports, Console, Power
Access Point (AP)	Provides Wi-Fi connectivity to wireless devices.	Ethernet port, Power port, Antennas (if present)

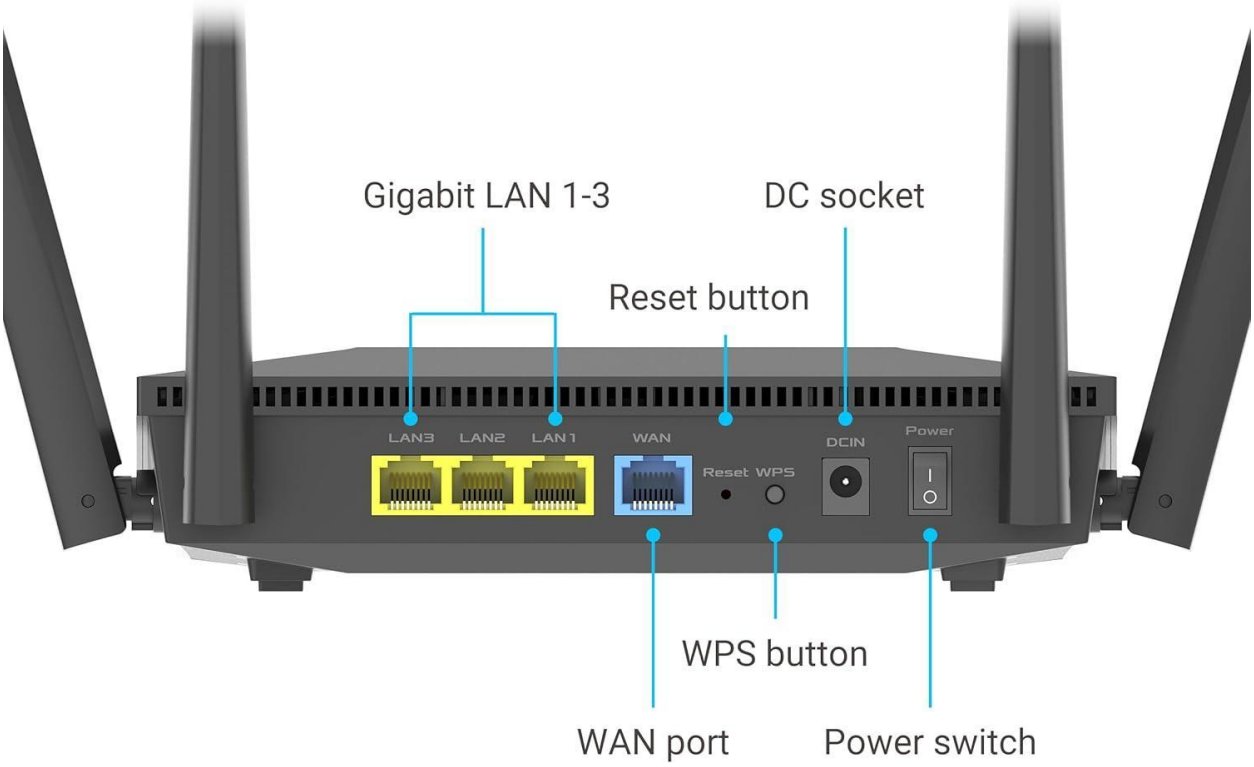
Note: TCP/UDP ports such as 80, 443, and 53 are different from physical hardware ports. The table lists physical ports.

2 Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.



A. Switch

- Ethernet ports — connect PCs and printers.
- Console port — configure the switch.
- Power port — supplies electricity.



B. Router

- WAN port — connects to the modem or internet line.
- LAN ports — connect wired devices.
- Power port — supplies electricity.
- Antennas — transmit and receive Wi-Fi signals.



C. Access Point

- Ethernet port — connects to a router or switch.
- Power port — supplies electricity, if separately powered.
- Antennas — send and receive wireless signals, if present.

3.

Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

- When I open Instagram on my smartphone, the access point connects my phone to the network through Wi-Fi.
- The router forwards my internet traffic toward Instagram's servers.
- The firewall filters network traffic according to its security rules.
- Instagram sends photos and videos back through the network to my smartphone.

4 Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

One possible reason for slow college Wi-Fi and frequent disconnections is that the access point (AP) cannot handle too many connected devices at the same time. In a college, many students use smartphones and laptops simultaneously to browse websites, attend online classes, and download files. This can overload the access point and cause slow internet speeds and unstable connections.

Solution: The college can upgrade its existing access point to a modern Wi-Fi 6 access point. Wi-Fi 6 can handle many connected devices more efficiently and improve network performance in crowded areas. The college should also check whether the switch has enough capacity to transfer data and whether the router can handle the total internet traffic. If the Wi-Fi signal is weak in some classrooms, installing additional access points in suitable locations can improve coverage.

Conclusion: Upgrading the access point and checking the capacity of the switch and router can improve Wi-Fi speed, reduce disconnections, and provide students with a more reliable internet connection.